Examining Digital Humanities Projects through the Lens of Technical and Professional Communication

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The field of technical and professional communication (TPC) explores topics that naturally overlap with digital humanities (DH.) However, this overlap has not been fully explored. This presentation argues that DH would benefit from additional research conducted through the lens of TPC. It also reports the findings of a study that uses TPC methods to examine the genre of DH project websites.

According to Carolyn Rude, the central research question in TPC is "how do texts (print, digital, multimedia; visual, verbal) and related communication practices mediate knowledge, values, and action in a variety of social and professional contexts?" (Rude 2009: 176). This question is typically explored by TPC scholars in one of four areas: "disciplinarity" (research about the field of TPC,) "pedagogy" (research about teaching in TPC,) "practice" (research about the creation and management of documents,) and "social change" (research about the changes documents make in the world) (Rude 2009: 176). TPC scholars who have examined TPC and DH overlaps have focused on pedagogy (Ballentine 2015: 1-3; Litterio 2021: 77-88; B. Ballentine, 2022: 24-37). However, the other three research areas could also be used to explore DH. This presentation describes one study in which TPC "practice" research is applied to DH.

TPC "practice" research examines the effective design of documents, and in recent decades, the term "documents" has expanded to include multimodal material, such as websites (Rude 2009: 175). Often effective document design involves using design features that are conventional to a particular genre of document, which helps the designer situate the document in relation to similar ones and streamlines people's use of the document (Bateman 2008: 177-248). This study examines document design within the genre of online DH projects by reporting the usage of 59 features in the interfaces of 100 DH projects. In doing so, it supports the development of a better understanding of the genre of DH projects, which in turn can be used to enhance their design.

Central to the study is the TPC research method of content analysis, which is defined in the field as "a research technique for making replicable and valid inferences from texts (or other meaningful matter) to the contexts of their use" (Krippendorff 2018: 24). For this study, I applied the four-step quantitative content analysis method developed by Ryan K. Boettger and Laura A. Palmer (Boettger / Palmer 2010: 346-357).

In the first step of this study, "identifying a sample," I gathered all of the DH projects listed in the "Project Registry, A-Z" page on the *Reviews in Digital Humanities* website. This step, which was conducted in November 2024, yielded a total of 219 projects. I then assigned each project a number and used a random number generator to select 100 projects for my study sample.

In the second step, "collecting and categorizing data," I developed a list of features to examine in all the sample projects. First, I explored existing literature to compile a list of common web design features in similar website genres. These genres included university, museum education, art museum library, cultural heritage, and special library websites (Hite / Railsback 2010: 109-111; Huffman 2011: 24-35; Manjunatha 2016: 3-7; Piñeiro-Naval et al. 2014: 659–665; Saiki 2010: 55-61). I developed a list of design features using emergent coding, a method in TPC content analysis in which noteworthy features "emerge" through observation (Thayer et al. 2007: 268). For my emergent coding, I randomly selected 10 DH projects from the 119 projects that were not included in the study sample, and I closely examined each of them. Finally, I selected 10 accessibility-specific features using emergent coding in combination with the Level A criteria in the Website Accessibility Content Guidelines 2.2 (WACG 2.2) (Campbell et al. 2023). Altogether, I compiled a list of 59 total features to examine in my sample of 100 DH projects.

In the third step, "developing a coding scheme and training raters," I defined the 59 features and used the definitions to identify their presence in the sample projects, several of which were no longer online.

Finally, in step four, "analyzing the data," I used descriptive statistics to determine how common the features were among the DH projects that currently have websites (n = 92.)

My results demonstrate commonalities as well as differences among the websites. About half of the homepages contain a guide for using the web-

site (50%, *n* = 46) or access to a database or digital collection (51%, *n* = 47.) Many—but not all—of the sites have a main menu (77%, *n* = 71) or a footer (67%, *n* = 62) and most sites adjust to different screen sizes (90%, *n* = 83.) In terms of site identity, some projects have their own logo (49%, *n* = 45) explain how to cite themselves (36%, *n* = 33) and use first-person language (52%, *n* = 48.) Most of the project sites contain photographs (80%, *n* = 74,) but virtual tours (7%, *n* = 6) and digital 3D models (3%, *n* = 3) are less common. Regarding accessibility of the projects' homepages, most are coded such that the language can be changed (77%, *n* = 71) and the page can be navigated using a keyboard (79%, *n* = 73,) but only 26% (*n* = 24) of the projects have suitable alternative text. I will present these statistics and my other findings alongside data regarding the DH projects' home institutions and countries.

This TPC content analysis is a starting point for identifying genre conventions among DH projects. Scholars can refer to these conventions as they design and refine their DH projects, particularly as they tackle challenges related to sustainability and accessibility. Future research may examine the impact of certain design features in terms of usability and ecological impact. By undertaking a content analysis of DH projects, this presentation demonstrates one application of TPC methods to DH research. It also argues that TPC research can and should be applied to DH because that would broaden the research questions and answers available to scholars in both fields.

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